

NI TB Genomic Surveillance

Outbreak Investigation Report

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About This Report

This report is produced by the Northern Ireland TB Genomic Surveillance platform using whole-genome sequencing (WGS) data and epidemiological case records. It is intended to support TB programme staff and public health investigators by providing genomic evidence for transmission clusters, drug-resistance profiles, and programme performance metrics. **This is a decision-support tool only — all findings must be reviewed and acted on by a qualified clinician or public health professional. No automated decisions are made.**

TB Genomics — Key Concepts

Concept	Explanation
Whole-Genome Sequencing (WGS)	Reads the complete ~4.4 Mb genome of <i>M. tuberculosis</i> . More informative than conventional typing (MIRU, spoligotyping).
SNP (single nucleotide polymorphism)	A single base-pair difference in the genome. Closely related strains share very few SNPs. Used as a genetic 'distance' metric.
SNP threshold for transmission	Strains with ≤12 SNPs are considered potentially linked (UK NICE guidance). ≤5 SNPs suggests recent direct transmission. >50 SNPs effectively rules out recent shared transmission.
Lineage	<i>M. tuberculosis</i> is classified into 7+ major lineages (L1–L7) reflecting global evolutionary history. Lineage influences drug-resistance patterns and may correlate with transmissibility.
Cluster	A group of cases whose sequences are genetically similar (within the SNP threshold). A cluster does not prove direct person-to-person transmission — epidemiological linkage is required to confirm transmission routes.
outbreaker2	A Bayesian MCMC method that combines SNP distances with collection dates and an assumed generation time to probabilistically infer who-infected-whom. Output posterior probabilities indicate the likelihood of a direct transmission event between any pair of cases.
Generation time	The average time between one person being infected and the next person they infect being detected. For TB, this is typically 1–3 years (range 0.5–5 years) due to the long latency period.
MCMC convergence	Markov Chain Monte Carlo simulations must reach a stable state ('converge'). A convergence diagnostic near 1.0 indicates reliable estimates. Values >1.1 suggest the chain has not fully mixed and results should be interpreted cautiously.
Drug resistance	Genomic mutations predict resistance to first-line drugs (isoniazid, rifampicin, etc.) and define MDR-TB (multi-drug resistant) and XDR-TB (extensively drug resistant). Genomic DR prediction is used alongside phenotypic DST.
Transmission network	A directed graph where arrows indicate the most probable direction of transmission. High-confidence links (posterior probability >0.70) warrant immediate epidemiological follow-up to confirm exposure history.

Table 1. TB genomics reference — key terms used throughout this report.

Case Summary

Total Cases	25
Clustered Cases	14
Unclustered Cases	11
Open Clusters	3

Table 2. Programme case count summary. Clustered cases are those linked by genomic similarity to at least one other case. Unclustered (singleton) cases may represent imported strains, sporadic transmission, or reactivation of latent disease. Open clusters are active genomic transmission clusters with ongoing epidemiological investigation.

Automated Interpretation Flags

- QC pass rate is below target: 60.0% (target $\geq 90\%$).
- Contamination flags detected: 1 cases require review.
- Open clusters requiring investigation: 3.
- High-confidence transmission links present; prioritize epidemiology follow-up.

Analysis Summary

Posterior Samples	1500
MCMC Iterations	2000
Burn-in	500
Mean Log-Likelihood	-18757.56
Log-Likelihood SD	3239.25

Table 3. outbreaker2 Bayesian MCMC analysis parameters. **Posterior Samples** is the number of accepted MCMC draws used to compute estimates — higher values give more stable posteriors. **Mean Log-Likelihood** reflects model fit; values closer to zero (less negative) indicate better fit. **Transmission Probability** is the average posterior probability that any given case-pair represents a direct transmission event. **Generation Time** is the modelled average interval (in days) between infection events in a transmission chain. **Convergence Diagnostic** near 1.0 confirms the MCMC chain has stabilised; values >1.1 indicate cautious interpretation is needed.

How to interpret outbreaker2 results: outbreaker2 reconstructs the most probable transmission tree using both genetic distance (SNPs) and timing (collection dates). Pairs with high posterior transmission probability (>0.5) represent genomically and temporally plausible direct transmission events. These are candidates for epidemiological investigation to identify shared exposure. Lower probability pairs may still be linked within the same cluster but through one or more undetected intermediate cases.

Programme Surveillance KPIs (Last 12 Weeks)

Eligible Cases	5
Sequenced Cases	5
Sequencing Coverage (%)	100.0
QC Reported Cases	5
QC Pass Cases	3
QC Fail Cases	2
QC Pass Rate (%)	60.0
Contamination Flags	1
Median Days Specimen to QC	8.44

Table 4. Programme surveillance KPIs over the reporting window. **Sequencing Coverage** is the percentage of eligible TB culture-confirmed cases that have received whole-genome sequencing. The UK target is $\geq 80\%$. **QC Pass Rate** is the percentage of sequenced samples that meet quality thresholds (e.g. $\geq 95\%$ genome coverage at $\geq 10\times$). Low pass rates may indicate DNA quality issues, contamination, or laboratory process variation. **Contamination Flags** indicate samples where a mixed-strain signal suggests cross-contamination requiring repeat or rejection.

Regional Sequencing Representativeness

Region	Eligible	Sequenced	Coverage %
United Kingdom	5	5	100.0

Table 5. Regional sequencing representativeness. Regions with coverage $<80\%$ may introduce ascertainment bias — clusters in under-sequenced regions may be underdetected. Where persistent regional gaps exist, review laboratory submission pathways and specimen transport processes.

Weekly Surveillance Trends (12 Weeks)

Week	Eligible	Sequenced	Coverage %	QC Pass %
2026-02-23	0	0	None	None
2026-03-02	1	1	100.00	0.00
2026-03-09	1	1	100.00	100.00
2026-03-16	0	0	None	None
2026-03-23	0	0	None	None
2026-03-30	0	0	None	None
2026-04-06	0	0	None	None
2026-04-13	0	0	None	None
2026-04-20	0	0	None	None
2026-04-27	1	1	100.00	100.00
2026-05-04	0	0	None	None
2026-05-11	0	0	None	None

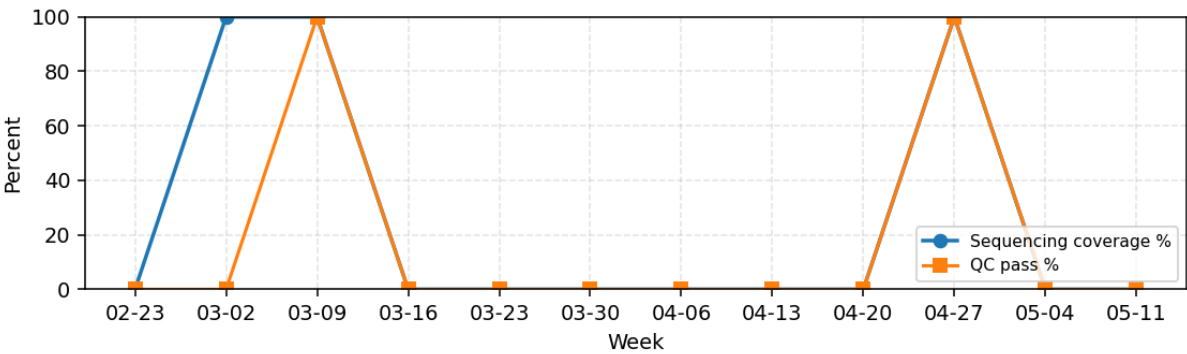


Table 6. Weekly sequencing coverage and QC pass rates over the last 12 weeks. Figure 1 (below) plots coverage and QC trends — a declining trend may indicate emerging laboratory issues. Weeks with zero eligible cases may reflect reporting lags rather than true absence of TB.

Cluster Action Prioritization

Cluster	Cases	Regions	Most Recent	Status	Priority
5cfd8109	6	1	2026-05-02	open	21
a2228458	6	1	2026-01-25	open	18
cc57094c	2	1	2026-01-05	open	10

Table 7. Clusters ranked by investigation priority score. Score is composite: cluster size (×2), cross-region spread (×3), specimen recency within 14/30/60 days (×3/2/1), open investigation status (×3). Higher scores indicate clusters warranting urgent epidemiological follow-up. **Status 'open'** means an active field investigation is ongoing or recommended.

Recommended action: For clusters with priority score >10 and status 'open', ensure field epidemiology is actively investigating shared exposure (household contacts, healthcare settings, social networks). Cross-region clusters may indicate transmission events during travel or care-seeking across NHS trust boundaries.

Lineage and Drug Resistance Validation

Interpreted Samples	25
Samples with Lineage	25
Lineage Coverage (%)	100.0
Samples with Resistance Calls	25
Resistance Coverage (%)	100.0
Any Resistance Signal	12
Rifampicin-Resistant (suspected)	12
Isoniazid-Resistant (suspected)	12
MDR (suspected)	12
FQ-Resistant (suspected)	12

Table 8. Lineage and drug-resistance epidemiology summary. Rows prioritize actionable burden indicators (coverage, resistance signal counts, and suspected RR/MDR/FQ resistance) to support triage and follow-up planning.

Top observed lineages: L4: 12, L2: 7, L3: 4, L1: 2

Secondary Transmission Engines

Consensus State	primary_evidence_only
Primary Network Available	True
High-Confidence Links	22
Priority Nodes Flagged	10
Cross-Method Precision	0.341
Cross-Method Recall	1.0
Cross-Method Jaccard	0.341

Table 9. Secondary transmission evidence summary. This section reports actionable signals from network inference and cross-method agreement, focusing on whether transmission hypotheses are strong enough to prioritize field investigation.

Action signal: high-confidence transmission links are present; prioritize contact tracing and exposure verification for flagged nodes.

Cross-Method Clustering Comparison

Sequence Assigned Cases	14
Outbreaker Assigned Cases	25
Overlap Cases	14
Pairwise Precision	0.341
Pairwise Recall	1.0
Pairwise Jaccard	0.341

Table 10. Agreement between SNP-threshold sequence clustering and outbreaker2 probabilistic clustering. **Precision**: of pairs grouped together by outbreaker2, the fraction also grouped by sequence clusters. **Recall**: of pairs grouped by sequence clusters, the fraction also grouped by outbreaker2. **Jaccard**: overall overlap index (0=no agreement, 1=perfect agreement). Discordant pairs — grouped by one method but not the other — may represent cases where temporal data (outbreaker2) overrides genomic distance alone, or where the SNP threshold is set differently.

Sequence Clustering Snapshot

Cluster Count	3
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Case-Level Operational Actions

Case	Cluster	SNP NN	Likely Link	Posterior	Resistance	QC	Recommended Action
2c6b4e7e	5cfd8109	25	9ced0378 -> 2c6b4e7e	1.000	susceptible	pass	Immediate contact tracing
59dfd74d	5cfd8109	25	9ced0378 -> 59dfd74d	1.000	susceptible	pass	Immediate contact tracing
403876ae	none	n/a	0ee523da -> 403876ae	0.625	pyrazinamide	fail	Repeat/verify sequence before action
9ced0378	5cfd8109	25	de2c3375 -> 9ced0378	1.000	isoniazid	pass	Immediate contact tracing
261b1032	none	n/a	67ae293e -> 261b1032	1.000	isoniazid	fail +contam	Repeat/verify sequence before action
af138d22	a2228458	25	a27bca47 -> af138d22	1.000	pyrazinamide	fail	Repeat/verify sequence before action
4c373fae	cc57094c	25	878eb85d -> 4c373fae	1.000	susceptible	fail	Repeat/verify sequence before action
a27bca47	a2228458	25	57f4935a -> a27bca47	1.000	susceptible	pass	Immediate contact tracing
eb37becb	none	n/a	de2c3375 -> eb37becb	1.000	susceptible	pass	Immediate contact tracing
67ae293e	none	n/a	67ae293e -> 261b1032	1.000	pyrazinamide, fluoroquinolones	fail	Repeat/verify sequence before action
57f4935a	a2228458	25	f4edc9e4 -> 57f4935a	1.000	susceptible	pass	Immediate contact tracing
f4edc9e4	a2228458	25	5f0b4f25 -> f4edc9e4	1.000	rifampicin	pass	Immediate contact tracing
de2c3375	5cfd8109	25	de2c3375 -> eb37becb	1.000	ethambutol	pass	Immediate contact tracing
1ac69380	none	n/a	878eb85d -> 1ac69380	0.977	susceptible	not_reported	Repeat/verify sequence before action
363c360d	a2228458	25	5f0b4f25 -> 363c360d	1.000	pyrazinamide, fluoroquinolones	fail	Repeat/verify sequence before action
5f0b4f25	a2228458	25	5f0b4f25 -> 363c360d	1.000	susceptible	pass	Immediate contact tracing
878eb85d	cc57094c	25	da73e225 -> 878eb85d	1.000	pyrazinamide	pass	Immediate contact tracing
750ef5d7	none	n/a	0ee523da -> 750ef5d7	0.995	susceptible	pass	Immediate contact tracing
5a363f94	5cfd8109	25	186f192b -> 5a363f94	1.000	susceptible	fail	Repeat/verify sequence before action
0ee523da	none	n/a	e28a8c42 -> 0ee523da	1.000	fluoroquinolones	not_reported	Repeat/verify sequence before action
86fc209c	none	n/a	0ee523da -> 86fc209c	1.000	susceptible	not_reported	Repeat/verify sequence before action
5e0b87ec	none	n/a	e28a8c42 -> 5e0b87ec	1.000	rifampicin	fail	Repeat/verify sequence before action
da73e225	none	n/a	e28a8c42 -> da73e225	1.000	rifampicin	not_reported	Repeat/verify sequence before action
186f192b	5cfd8109	25	186f192b -> 5a363f94	1.000	susceptible	pass	Immediate contact tracing
e28a8c42	none	n/a	e28a8c42 -> da73e225	1.000	susceptible	not_reported	Repeat/verify sequence before action

Table 11. Case-level operational action table linking genomic evidence to immediate field actions. SNP NN uses an available cluster SNP proxy where pairwise SNP matrices are not available; actions remain decision-support and require public-health review.

High-Confidence Transmission Pairs

Source	Recipient	Posterior	SNP Dist	Date Gap (d)	Shared Cluster	Action
e28a8c42	da73e225	1.000	n/a	27	no	Immediate trace + exposure review
e28a8c42	5e0b87ec	1.000	n/a	36	no	Immediate trace + exposure review
0ee523da	86fc209c	1.000	n/a	0	no	Immediate trace + exposure review
e28a8c42	0ee523da	1.000	n/a	52	no	Immediate trace + exposure review
186f192b	5a363f94	1.000	25	49	5cfd8109	Immediate trace + exposure review
da73e225	878eb85d	1.000	n/a	43	no	Immediate trace + exposure review
5f0b4f25	363c360d	1.000	25	23	a2228458	Immediate trace + exposure review
5f0b4f25	f4edc9e4	1.000	25	137	a2228458	Immediate trace + exposure review
f4edc9e4	57f4935a	1.000	25	93	a2228458	Immediate trace + exposure review
de2c3375	eb37becb	1.000	n/a	178	no	Immediate trace + exposure review
57f4935a	a27bca47	1.000	25	56	a2228458	Immediate trace + exposure review
878eb85d	4c373fae	1.000	25	341	cc57094c	Immediate trace + exposure review
a27bca47	af138d22	1.000	25	25	a2228458	Immediate trace + exposure review
67ae293e	261b1032	1.000	n/a	92	no	Immediate trace + exposure review
de2c3375	9ced0378	1.000	25	269	5cfd8109	Immediate trace + exposure review
9ced0378	59dfd74d	1.000	25	17	5cfd8109	Immediate trace + exposure review
9ced0378	2c6b4e7e	1.000	25	69	5cfd8109	Immediate trace + exposure review
0ee523da	186f192b	0.999	n/a	44	no	Immediate trace + exposure review
0ee523da	750ef5d7	0.995	n/a	12	no	Immediate trace + exposure review
0ee523da	67ae293e	0.989	n/a	311	no	Immediate trace + exposure review
878eb85d	1ac69380	0.977	n/a	84	no	Immediate trace + exposure review
5a363f94	de2c3375	0.975	25	133	5cfd8109	Immediate trace + exposure review
186f192b	5f0b4f25	0.776	n/a	112	no	Review with cluster context

Table 12. High-confidence source-recipient pairs (posterior ≥ 0.70) for contact-tracing prioritization, to be interpreted with epidemiological corroboration.

Discordant Method Review

Case Pair	SNP Result	Outbreaker2	SNP Dist	Posterior	Interpretation
de2c3375-eb37becb	not_clustered	linked	n/a	1.000	Temporal support without SNP-threshold support; assess missing in
da73e225-e28a8c42	not_clustered	linked	n/a	1.000	Temporal support without SNP-threshold support; assess missing in
0ee523da-86fc209c	not_clustered	linked	n/a	1.000	Temporal support without SNP-threshold support; assess missing in
5e0b87ec-e28a8c42	not_clustered	linked	n/a	1.000	Temporal support without SNP-threshold support; assess missing in
0ee523da-e28a8c42	not_clustered	linked	n/a	1.000	Temporal support without SNP-threshold support; assess missing in
878eb85d-da73e225	not_clustered	linked	n/a	1.000	Temporal support without SNP-threshold support; assess missing in
261b1032-67ae293e	not_clustered	linked	n/a	1.000	Temporal support without SNP-threshold support; assess missing in
0ee523da-186f192b	not_clustered	linked	n/a	0.999	Temporal support without SNP-threshold support; assess missing in
0ee523da-750ef5d7	not_clustered	linked	n/a	0.995	Temporal support without SNP-threshold support; assess missing in
0ee523da-67ae293e	not_clustered	linked	n/a	0.989	Temporal support without SNP-threshold support; assess missing in
1ac69380-878eb85d	not_clustered	linked	n/a	0.977	Temporal support without SNP-threshold support; assess missing in
186f192b-5f0b4f25	not_clustered	linked	n/a	0.776	Temporal support without SNP-threshold support; assess missing in
0ee523da-403876ae	not_clustered	linked	n/a	0.625	Temporal support without SNP-threshold support; assess missing in
363c360d-57f4935a	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
186f192b-de2c3375	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
a27bca47-f4edc9e4	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
186f192b-9ced0378	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
363c360d-a27bca47	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
57f4935a-af138d22	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
5f0b4f25-a27bca47	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
363c360d-af138d22	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
59dfd74d-de2c3375	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
5a363f94-9ced0378	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
57f4935a-5f0b4f25	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
5f0b4f25-af138d22	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
2c6b4e7e-5a363f94	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
186f192b-2c6b4e7e	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
2c6b4e7e-59dfd74d	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
186f192b-59dfd74d	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
2c6b4e7e-de2c3375	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
59dfd74d-5a363f94	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
363c360d-f4edc9e4	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh
af138d22-f4edc9e4	same_cluster	not_linked	25	n/a	SNP-clustered but no directed outbreaker support; possible older/sh

Table 13. Discordant pairs between SNP-threshold clustering and outbreaker2 linkage to support structured multidisciplinary adjudication.

QC Failure Drill-Down

Sample	QC	Coverage	Depth	Contam	Mixed-Lineage	Repeat Required
403876ae	fail	92.82	64.8	no	no	yes
261b1032	fail	92.9	264.4	yes	no	yes
af138d22	fail	93.93	173.4	no	no	yes
4c373fae	fail	94.17	197.5	no	no	yes
67ae293e	fail	93.56	151.6	no	no	yes
1ac69380	not_reported	n/a	n/a	no	no	yes
363c360d	fail	94.61	89.5	no	no	yes
5a363f94	fail	92.23	275.3	no	no	yes
0ee523da	not_reported	n/a	n/a	no	no	yes
86fc209c	not_reported	n/a	n/a	no	no	yes
5e0b87ec	fail	92.66	73.4	no	no	yes
da73e225	not_reported	n/a	n/a	no	no	yes
e28a8c42	not_reported	n/a	n/a	no	no	yes

Table 14. QC drill-down identifying samples requiring repeat testing or cautious interpretation before operational decisions.

Drug-Resistance Mutation Details

Case	Drug	Mutation	Gene	Confidence	Predicted	Phenotypic DST
0ee523da	fluoroquinolones	H526Y	gyrA	n/a	fluoroquinolones	not_available
261b1032	isoniazid	S531L	gyrA	n/a	isoniazid	not_available
363c360d	pyrazinamide	C15T	gyrA	n/a	pyrazinamide, fluoroquinolones	not_available
363c360d	fluoroquinolones	S531L	embB	n/a	pyrazinamide, fluoroquinolones	not_available
403876ae	pyrazinamide	D516V	gyrA	n/a	pyrazinamide	not_available
5e0b87ec	rifampicin	H526Y	gyrA	n/a	rifampicin	not_available
67ae293e	pyrazinamide	Q431K	katG	n/a	pyrazinamide, fluoroquinolones	not_available
67ae293e	fluoroquinolones	C15T	inhA	n/a	pyrazinamide, fluoroquinolones	not_available
878eb85d	pyrazinamide	Q431K	rpoB	n/a	pyrazinamide	not_available
9ced0378	isoniazid	H526Y	inhA	n/a	isoniazid	not_available
af138d22	pyrazinamide	S531L	gyrA	n/a	pyrazinamide	not_available
da73e225	rifampicin	Q431K	rpoB	n/a	rifampicin	not_available
de2c3375	ethambutol	S315T	pncA	n/a	ethambutol	not_available
f4edc9e4	rifampicin	C15T	pncA	n/a	rifampicin	not_available

Table 15. Drug-resistance mutation evidence supporting predicted phenotype. Phenotypic DST linkage should be incorporated when an accredited laboratory DST feed is available.

Cluster Epidemiology Summary

Cluster	Cases	First	Latest	Median SNP	Max SNP	RR/MDR	Index	Recent(30/60/90)	Status
5cfd8109	6	2024-11-28	2026-05-02	25.0	25	2/2	186f192b	1/1/3	open+linked
a2228458	6	2025-03-20	2026-01-25	25.0	25	3/3	5f0b4f25	0/0/0	open+linked
cc57094c	2	2025-01-29	2026-01-05	25.0	25	1/1	878eb85d	0/0/0	open+linked

Table 16. Cluster-level epidemiology profile including growth recency and RR/MDR burden (RR/MDR reported as case counts).

Geographical Cluster Spread

Cluster	Regions	Cases	Cross-Region	Risk Flag
5cfd8109	United Kingdom	6	no	high
a2228458	United Kingdom	6	no	high
cc57094c	United Kingdom	2	no	high

Table 17. Cluster geography summary at region/trust-safe level (no household-level identifiers displayed).

Missing-Data Dashboard

Field	Completeness	Missing	Impact
Collection date	100.0%	0	affects outbreaker2 timing
Notification date (proxy)	100.0%	0	affects surveillance timeline
Region/trust	100.0%	0	affects representativeness and spread
Lineage	100.0%	0	affects phylogenetic interpretation
Resistance call	100.0%	0	affects clinical action
QC metrics	80.0%	5	affects confidence in inference

Table 18. Completeness and impact dashboard for key outbreak-investigation data fields, aligned to transparent inference limitations.

Model Reliability Diagnostics

Diagnostic	Value	Threshold	Interpretation
Effective sample size	n/a	>200 (rule-of-thumb)	Higher is more stable
Gelman-Rubin / R-hat	n/a	<1.1	Convergence confidence
Acceptance rate	n/a	0.2-0.6 (typical)	Mixing efficiency
Number of chains	n/a	>=2	Multi-chain robustness
Sensitivity run performed	no	yes	Parameter robustness
Overall reliability	exploratory		Use exploratory language if not converged

Table 19. Model-diagnostic reliability section distinguishing operationally reliable vs exploratory outbreaker2 inference.

Transmission Routes Reference

Understanding TB Transmission Routes from WGS

Whole-genome sequencing identifies genomic relatedness but does not directly observe contact events. Combining genomic clusters with epidemiological data (contact tracing, shared locations, timeline of diagnosis) allows investigators to build a plausible transmission chain. The table below summarises the genomic signals and their transmission implications.

Genomic Signal	SNP Range	Transmission Implication	Recommended Action
Highly probable direct transmission	0–5 SNPs	Strong genomic evidence of recent direct person-to-person transmission. Strain has had little time to evolve.	Immediate contact tracing; identify shared setting (household, workplace, healthcare).
Possible direct or near-direct transmission	6–12 SNPs	Genetically close; consistent with transmission within the last 1–3 years or via an undetected intermediate.	Epidemiological linkage investigation; check whether cases share contacts or settings.

Within extended cluster — indirect link likely	13–50 SNPs	Genetically related but too diverged for recent direct transmission. Likely share a common ancestor strain.	Review cluster history; may represent reactivation from the same source years earlier.
Unrelated strains	>50 SNPs	No plausible genomic link. Coincident diagnoses are likely due to independent exposure or reactivation.	No cluster-based action; manage as separate cases.
Mixed-lineage / contamination	N/A	Two or more distinct strain signals in a single sample. May indicate laboratory cross-contamination or mixed infection.	Flag for repeat sequencing; do not use in cluster assignments until resolved.

Table 20. TB transmission route classification by SNP distance (M. tuberculosis whole-genome comparison). SNP thresholds follow UK NICE guideline NG33 and published literature (Walker et al. 2013, Meehan et al. 2019). The generation-time prior used in outbreaker2 is programme-configured and influences whether a given SNP distance is interpreted as consistent with direct versus indirect transmission.

Transmission Priority Signals

Case	Region	Risk	Out	In
0ee523da		3.5251	5	1
e28a8c42		2.1	3	0
9ced0378		1.7	2	1
de2c3375		1.6924	2	1
878eb85d		1.6837	2	1
5f0b4f25		1.6328	2	1
186f192b		1.5428	2	1
da73e225		1	1	1
f4edc9e4		1	1	1
57f4935a		1	1	1

Table 21. Top-priority cases by network centrality. Network snapshot: 25 nodes, 24 directed links, 22 high-confidence links (posterior >0.70). **Out** = outgoing transmission links (potential sources); **In** = incoming links (potential recipients). Cases with multiple outgoing high-confidence links ('superspreaders') should be prioritised for epidemiological investigation.

High-confidence transmission links (posterior probability >0.70) represent the strongest genomic evidence of direct transmission between a pair of cases. Investigators should review the contact history for all such pairs. Where epidemiological linkage can be confirmed, the direction of transmission (source → recipient) inferred by outbreaker2 can inform contact prioritisation for LTBI screening.

Diagnostic Graphics

The following plots are generated by outbreaker2 and the platform's supplementary visualisation pipeline. Each figure caption explains the content and how to interpret the output.

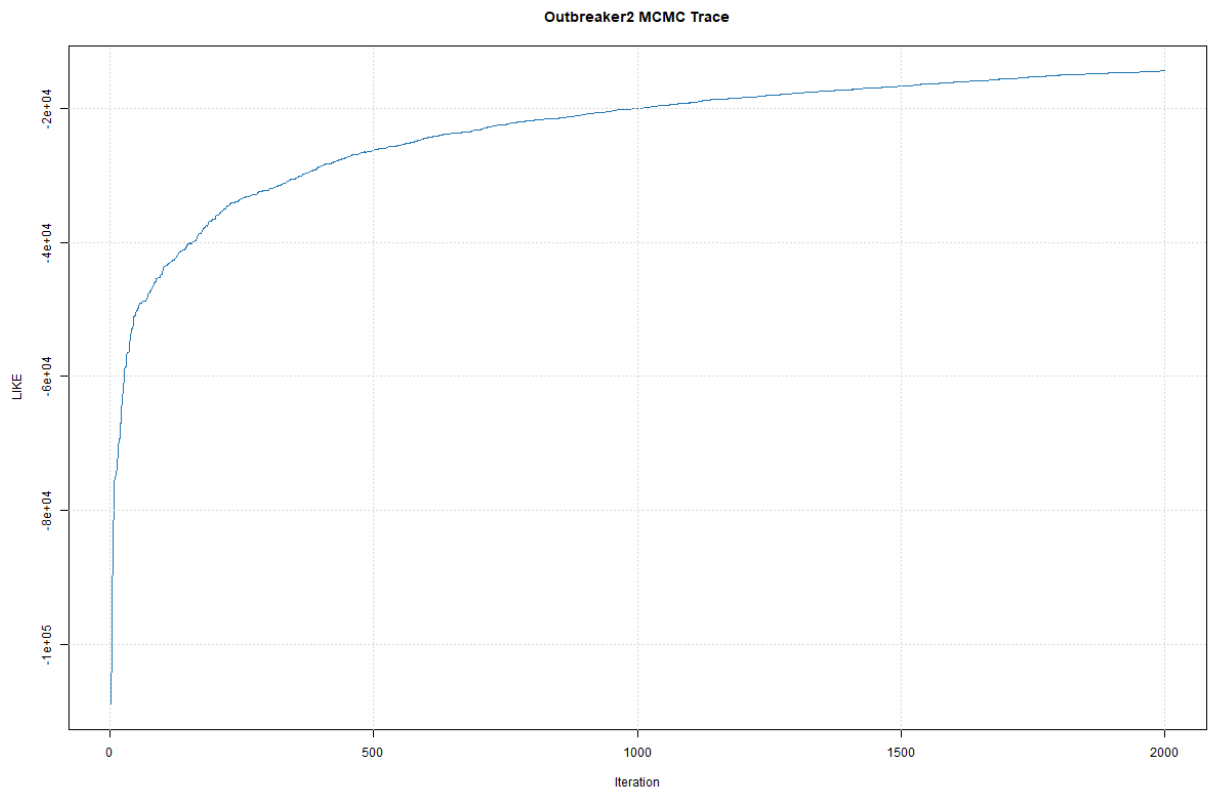


Figure 2. MCMC trace plot — log-posterior probability over iterations. A well-mixed chain shows stable fluctuation around a mean value (no upward/downward drift). If the trace shows a long burn-in slope or multiple plateaux, the chain may not have converged; consider increasing iterations or checking input data quality.

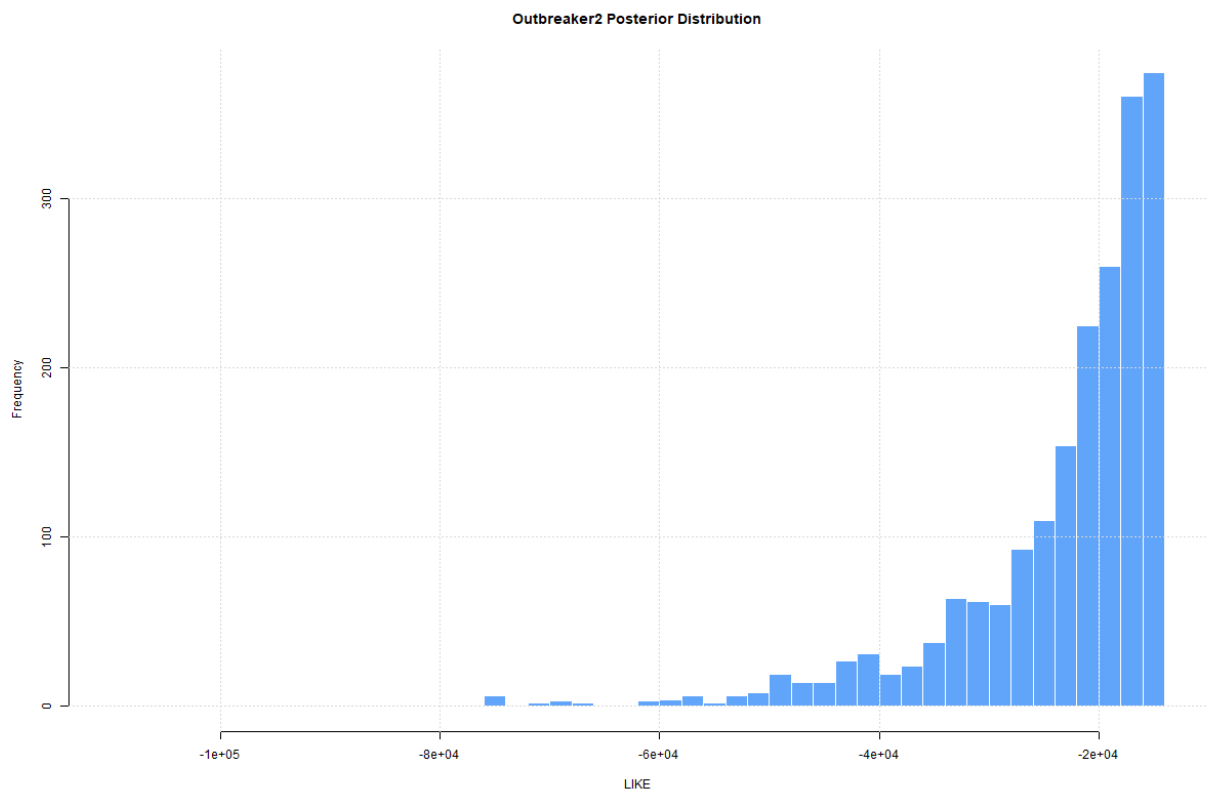


Figure 3. Posterior distribution histograms — marginal distributions of key model parameters (transmission probability, sampling probability, generation time). Narrow, symmetric peaks indicate well-determined parameters. Broad or multi-modal distributions suggest parameter uncertainty, which should be reflected in cautious interpretation of individual transmission links.

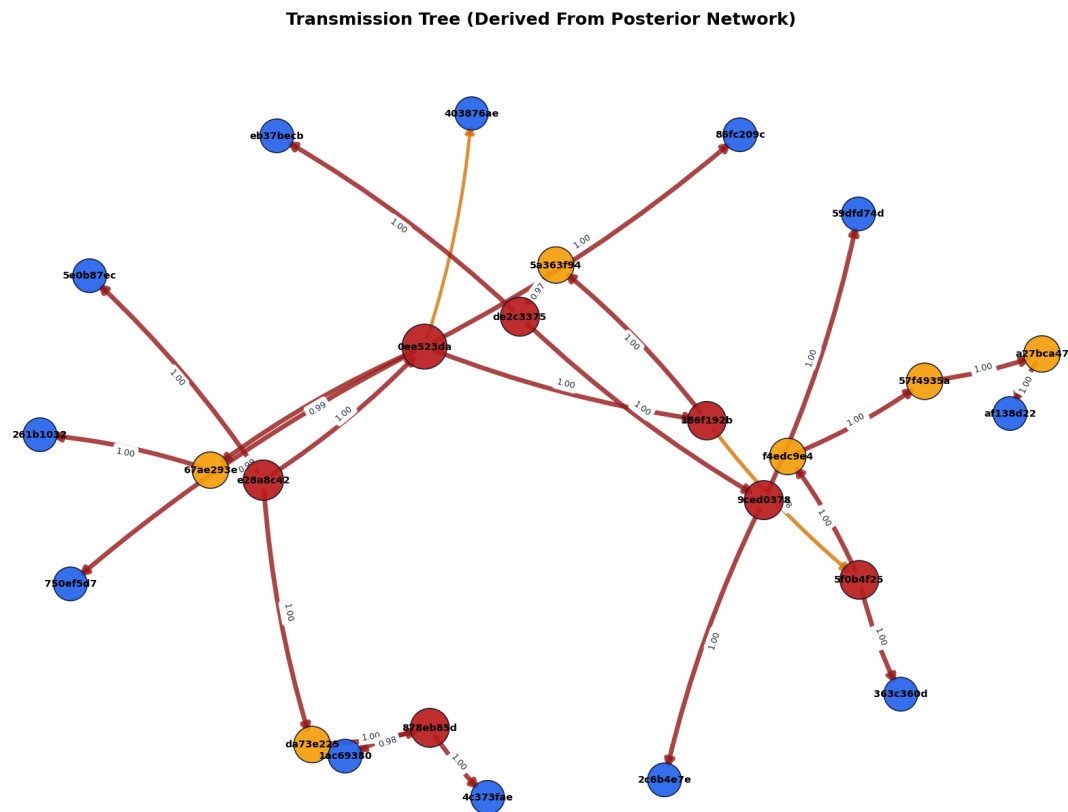


Figure 4. Inferred transmission tree (most probable who-infected-whom). Each node is a case; arrows indicate the direction of inferred transmission from source (tail) to recipient (head). Arrow thickness or colour (where shown) reflects posterior probability. Dashed or thin arrows indicate lower-confidence links. Cases with no incoming arrow are probable index cases or represent undetected importation events.

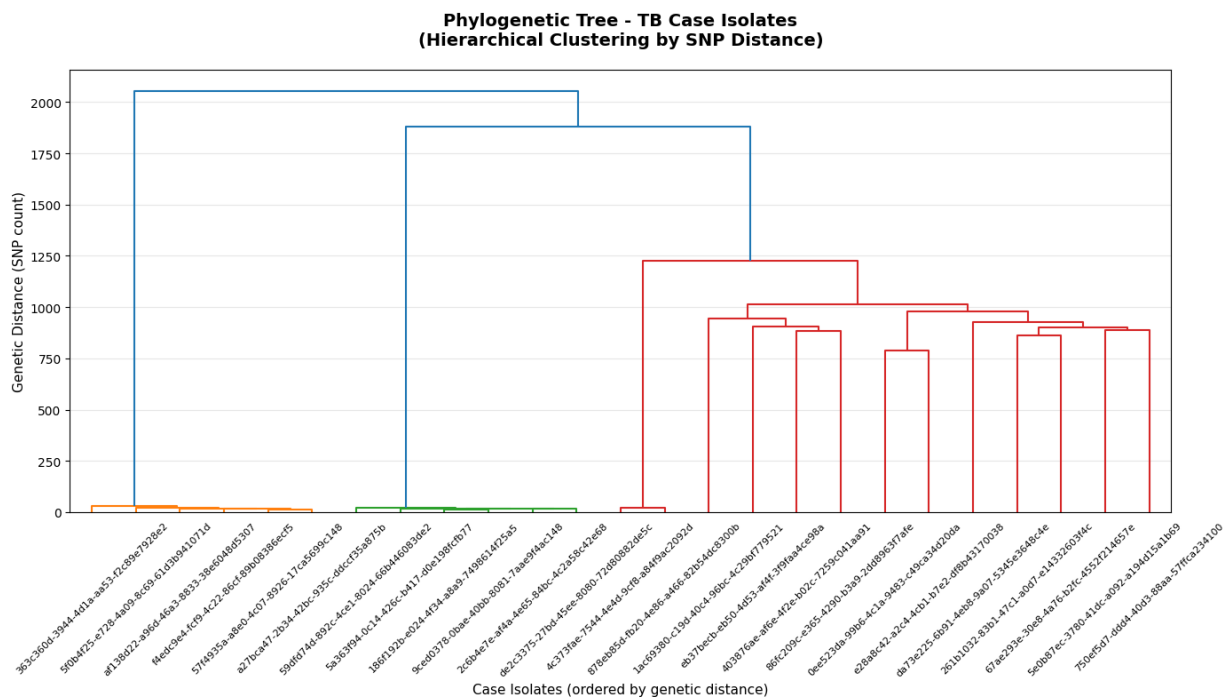


Figure 5. Phylogenetic context — a midpoint-rooted maximum parsimony or neighbour-joining tree of sequenced cases, coloured by cluster or region. Branch length represents SNP distance. Cases on short branches with few SNPs between them form tight clades consistent with recent transmission. Well-separated clades indicate genetically distinct strain lineages circulating concurrently.

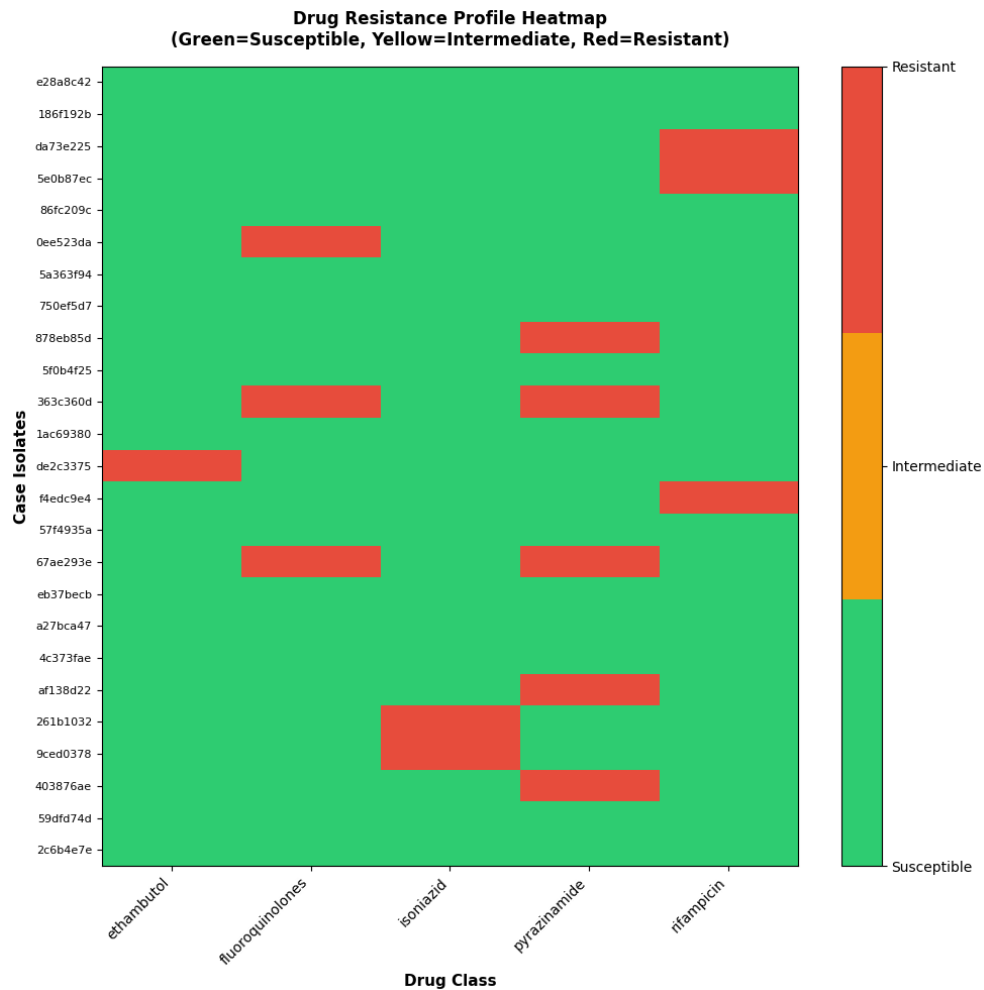


Figure 6. Drug resistance profile summary — frequency of predicted resistance mutations across the sequenced cohort. Bars represent the proportion of cases with predicted resistance to each antibiotic class. Rifampicin + isoniazid co-resistance defines MDR-TB. High frequencies of any first-line resistance warrant urgent review of empirical treatment protocols.

M. tuberculosis Lineage Reference

Lineage classification places a strain within the global phylogeny of *M. tuberculosis*. Lineage influences drug-resistance acquisition patterns and geographic origin. The following table provides clinical and epidemiological context for lineages commonly observed in Northern Ireland.

Lineage	Name / Origin	DR Association	NI Relevance	Notes
L1	East-African-Indian / Indo-Oceanic	Lower MDR-TB frequency	Cases linked to South Asia, Horn of Africa	Commonly found in Bangladeshi, Indian, Somali communities.
L2	East-Asian (Beijing lineage)	High MDR/XDR-TB risk; associated with resistance acquisition	Sporadic importation; watch for resistance	Beijing strains have shown high transmissibility in some outbreak settings.
L3	East-African-Indian (Delhi/CAS)	Moderate DR frequency	Cases linked to Pakistan, Afghanistan, India	Common in large urban TB programmes in the UK.
L4	Euro-American	Generally lower DR; but historical MDR clusters exist	Dominant UK-born strain type	Most legacy UK TB is L4. Reactivation common in older cohorts.
L5 / L6	West-African (<i>Mycobacterium africanum</i>)	Lower overall DR	Cases linked to West Africa	Slower growth, may present with atypical features.
L7	Ethiopian	Limited data	Rare in NI	Emerging lineage classification; limited clinical guidance available.

Table 13. *M. tuberculosis* lineage reference for clinical and epidemiological context. DR = drug resistance; MDR = multidrug-resistant; XDR = extensively drug-resistant. Lineage assignment should be combined with phenotypic DST and clinical judgment. Source: Coll et al. (2014) Nature Genetics; WHO Global TB Report 2023.

Clinical and Public Health Action Summary

The table below maps genomic findings to recommended clinical and public health actions. All actions must be confirmed by the responsible clinician and public health team.

Finding	Recommended Action	Urgency
New case links to an existing open cluster (≤ 12 SNPs)	Notify cluster lead; extend contact tracing to include new case contacts; review whether the cluster source has been identified.	Within 5 working days
High-confidence transmission link identified (posterior >0.70)	Epidemiological review of both cases; document shared exposure if found; update cluster investigation record.	Within 5 working days
New cluster opened (≥ 2 cases genetically linked, no prior cluster)	Open investigation; notify public health; assign epidemiologist; initiate contact tracing for all cases.	Within 2 working days
MDR-TB predicted by genomics	Confirm with phenotypic DST; notify MDR-TB specialist centre; initiate enhanced infection control if hospitalised.	Immediately on result
Sequencing coverage $<80\%$ for a region	Review specimen submission and transport processes for that region; identify cases that did not receive WGS and arrange retrospective sequencing if available.	Monthly programme review
MCMC convergence diagnostic >1.1	Do not rely on posterior transmission probabilities from this run. Increase MCMC iterations and re-run outbreaker2. Check input data completeness.	Before using results
Contamination flag on a sample	Exclude sample from cluster assignments; arrange repeat sequencing from original culture. Investigate laboratory process if multiple consecutive contamination flags.	Within 10 working days

Table 14. Recommended clinical and public health actions mapped to genomic findings. Urgency thresholds align with PHE/PHA TB operational guidance. All genomic findings must be reviewed in conjunction with clinical history, contact tracing records, and microbiological DST before action is taken.

Disclaimer: Genomic cluster assignments and transmission inferences are probabilistic estimates based on mathematical models. They supplement but do not replace epidemiological investigation. Do not use genomic evidence alone to assign legal or clinical liability for transmission.

Data Provenance

This report combines outbreaker outputs with surveillance KPIs, lineage/DR validation, secondary engine readiness, and cross-method clustering comparison artifacts available at generation time.

Outbreaker summary timestamp: 2026-05-11T10:33:31Z

Transmission network timestamp: 2026-05-11T10:33:31Z

Lineage/DR validation timestamp: 2026-05-11T10:32:41.793180+00:00

Secondary engine validation timestamp: 2026-05-11T10:25:04Z

Method comparison timestamp: 2026-05-11T10:33:37.423897+00:00